Exam Date & Time: 29-Nov-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

Biomaterial-characterization techniques [BME 4052]

Marks: 50 **Duration: 180 mins.**

Descriptive

Answer all the questions. Section Duration: 180 mins

Answer all questions.
Assume any missing data with proper reasoning.

1A) Explain the different methods adopted to reduce surface energy in materials with appropriate illustration. (4)

1B) Discuss the primary types of aberrations in optical microscopy. (2)

1C) In X-ray diffraction (XRD) characterization of nanomaterial, selecting the appropriate anode material is crucial. Molybdenum emits X-rays with a wavelength of about 0.7107 Å, whereas Copper emits at roughly 1.5406 Å. Evaluate which anode material provides higher-resolution measurements and explain your choice. Discuss the how these X-rays are produced. (4)

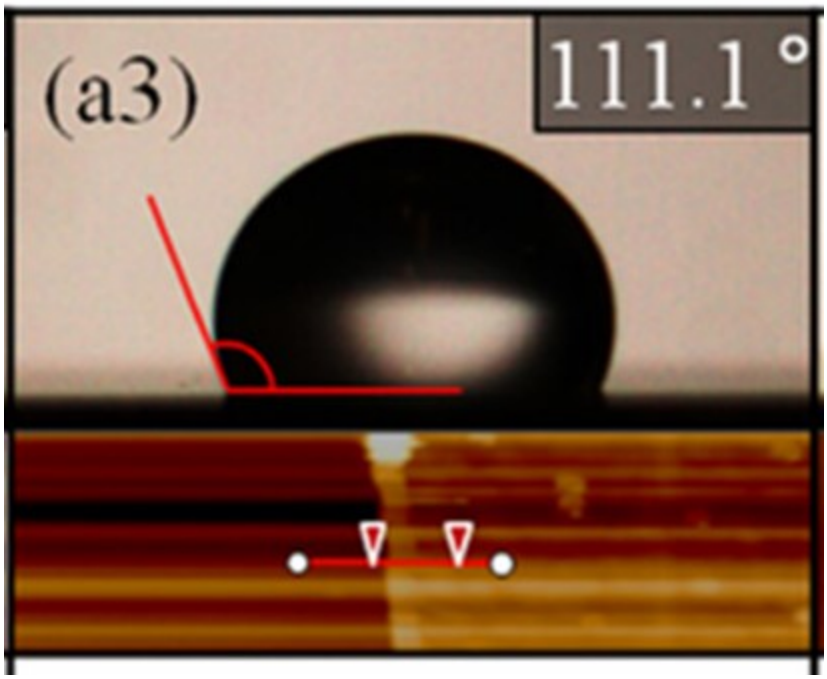
2A) Fatima characterized a nanomaterial using X-ray Diffraction (XRD). The following table shows the data she acquired. Calculate the average size of the nanoparticle using the data provided. Assume the X-ray wavelength $\lambda=1.54060A^{\circ}$ (for CuKα1 radiation) and the dimensionless shape factor K=0.94K. (3)

2θ (degrees)	**FWHM (degrees)**
29.54	0.210
36.75	0.225
42.90	0.235

2B) A research team is using a UV spectrophotometer to analyse the concentration of a protein sample that absorbs light at 280 nm. Given an absorbance reading of 0.65 at 280 nm with a path length of 1 cm and a molar absorptivity of 4.0×103 L/mol/cm. Calculate the concentration of the protein. Explain the working parts of UV spectrophotometer with suitable illustrations. (3)

2C) Explain the role of the Michelson interferometer in FTIR with suitable diagrams and describe how it generates an interferogram. (4)

3A) Trevor modified a surface with a polymer coating. Trevor characterized the surface using contact angle measurement and the results are shown in the below figure. Analyze and interpret the image given below (3)

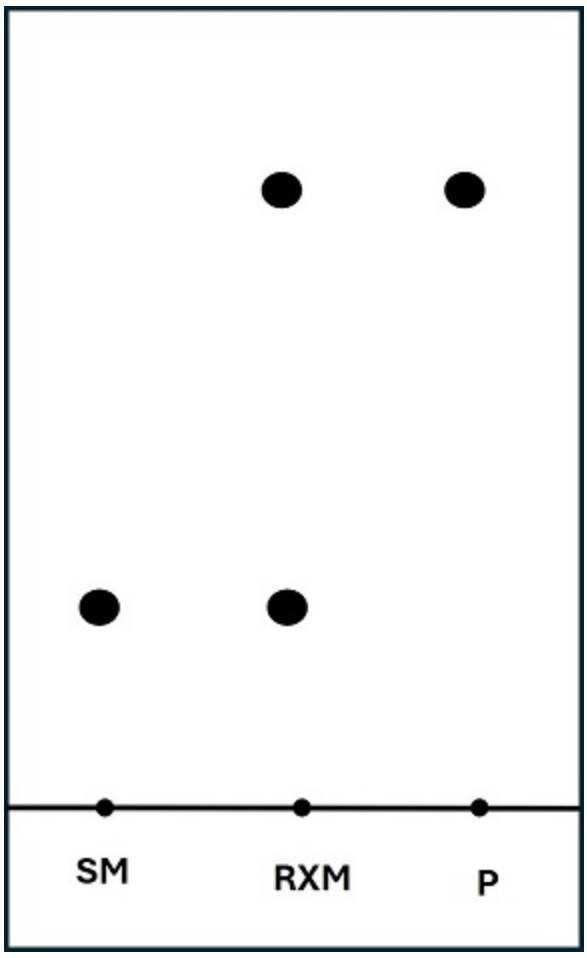


3B) Given a particle suspended in a solvent with a temperature of 298 K, and the viscosity of the solvent is 0.89 cP, Calculate the hydrodynamic radius (r) of a particle if its diffusion coefficient (D) is measured to be 4.0×10^{-13} m^2/s. (3)

3C) Two colloidal suspensions are stabilized differently: one with ionic surfactants showing a zeta potential of -45 mV, and another with polymeric stabilizers showing -12 mV. Analyze how these zeta potential differences influence the stability of each suspension. (4)

4A) Explain the principle and components of High-Performance Liquid Chromatography (HPLC) with relevant illustrations. (4)

4B) A chemist is monitoring a reaction's progress using Thin Layer Chromatography (TLC). The TLC plate shows three lanes labeled SM (Starting Material), RXN (Reaction Mixture), and P (Product). Analyze and interpret the TLC results. Based on the spot patterns in each lane, what conclusions can you draw about the reaction's progress and the presence of any unreacted starting material? (3)



4C)	Matrix-Assisted Laser Desorption/Ionization-Time of Flight (MALDI-TOF) mass spectrometry is widely used in proteomics for protein identification and characterization. Describe how MALDI-TOF aids in analysing complex proteins.	(3)
5A)	What is the rationale behind using an ion gun in X-ray Photoelectron Spectroscopy (XPS)?	(3)
5B)	In a Thermogravimetric Analysis (TGA) of a polymer, weight loss occurs in three steps: 50-150°C, 250-350°C, and 500-600°C. Analyze the thermal events responsible for each weight loss stage and discuss how the TGA data can help deduce the polymer's composition, thermal stability, and the presence of additives or residuals.	(4)
5C)	Explain how Time-of-Flight Secondary Ion Mass Spectrometry (TOF-SIMS) is used to analyze biomaterial surfaces and its role in assessing surface modifications.	(3)

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